

Single Port BRAM

//Verilog code

`timescale 1ns / 1ps

module singleport_BRAM(

input clk, we, [9:0] addr, [31:0] din, output [31:0] dout

);

reg [31:0]mem [0:1023];

always @(posedge clk)

begin

if(we) begin

mem[addr] <= din;

end

end

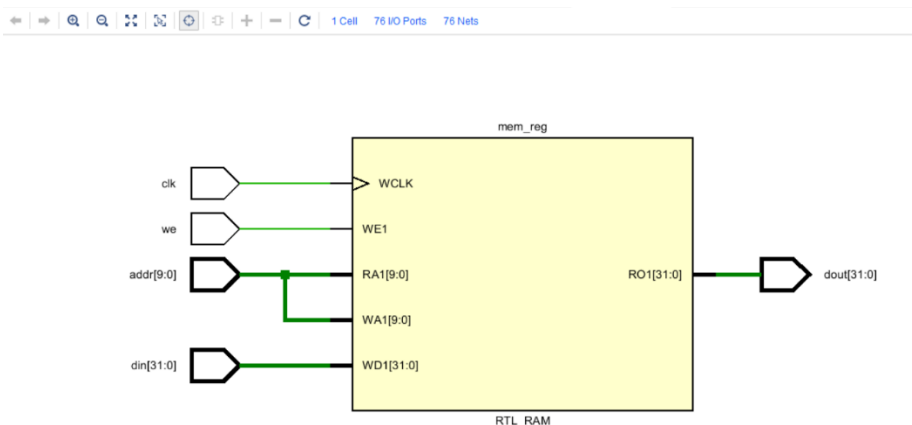
assign dout = mem[addr];

endmodule

1. Slice Logic

Site Type	Used	Fixed	Available	Util%
Slice LUTs*	548	0	53200	1.03
LUT as Logic	36	0	53200	0.07
LUT as Memory	512	0	17400	2.94
LUT as Distributed RAM	512	0		
LUT as Shift Register	0	0		
Slice Registers	0	0	106400	0.00
Register as Flip Flop	0	0	106400	0.00
Register as Latch	0	0	106400	0.00
F7 Muxes	256	0	26600	0.96
F8 Muxes	128	0	13300	0.96

RTL analysis schematic:



Look Up Tables:

Cell Properties			
mem_reg_768_1023_0_0_i_1			
I2	I1	I0	O=I0 & I1 & I2
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Table 1: LUT3

Cell Properties						
dout_OBUF[9]_inst_i_1						
I5	I4	I3	I2	I1	I0	O=I1 & I2 & !I4 + I0 & I2 & I4 + !I2 & I3 & I4 + !I2 & !I4 & I5
0	0	0	0	0	0	0
0	0	0	0	0	1	0
0	0	0	0	1	0	0
0	0	0	0	1	1	0
0	0	0	1	0	0	0
0	0	0	1	0	1	0
0	0	0	1	1	0	1
0	0	0	1	1	1	1
0	0	1	0	0	0	0
0	0	1	0	0	1	0
0	0	1	0	1	0	0
0	0	1	0	1	1	0
0	0	1	1	0	0	0
0	0	1	1	0	1	0
0	0	1	1	1	0	1
0	0	1	1	1	1	1
0	1	0	0	0	0	0
0	1	0	0	0	1	0
0	1	0	0	1	0	0
0	1	0	0	1	1	0
0	1	0	1	0	0	0
0	1	0	1	0	1	1
0	1	0	1	1	0	0
0	1	0	1	1	1	1
0	1	1	0	0	0	1
0	1	1	0	0	1	1
0	1	1	0	1	0	1
0	1	1	0	1	1	1
0	1	1	1	0	0	0
0	1	1	1	0	1	1
0	1	1	1	1	0	0
0	1	1	1	1	1	1
1	0	0	0	0	0	1
1	0	0	0	0	1	1
1	0	0	0	1	0	1
1	0	0	0	1	1	1
1	0	0	1	0	0	0
1	0	0	1	0	1	0
1	0	0	1	1	0	1
1	0	0	1	1	1	1
1	0	1	0	0	0	1
1	0	1	0	0	1	1
1	0	1	0	1	0	1
1	0	1	0	1	1	1
1	0	1	1	0	0	0
1	0	1	1	0	1	0
1	0	1	1	1	0	1
1	0	1	1	1	1	1
1	1	0	0	0	0	0
1	1	0	0	0	1	0
1	1	0	0	1	0	0
1	1	0	0	1	1	0
1	1	0	1	0	0	0
1	1	0	1	0	1	1
1	1	0	1	1	0	0
1	1	0	1	1	1	1
1	1	1	0	0	0	1
1	1	1	0	0	1	1
1	1	1	0	1	0	1
1	1	1	0	1	1	1
1	1	1	1	0	0	0
1	1	1	1	0	1	1
1	1	1	1	1	0	0
1	1	1	1	1	1	1

Table 2: LUT6

Single Port BRAM (Parameterized)

//Verilog code

`timescale 1ns / 1ps

module singlePort_para #(

parameter WIDTH = 32,

parameter DEPTH = 1024,

parameter ADDR_WIDTH = 10)

(input clk, we, [ADDR_WIDTH-1 :0] addr, [WIDTH-1 :0] din, output [WIDTH-1:0] dout

);

reg [WIDTH-1:0]mem [0:DEPTH-1];

always @(posedge clk)

begin

if(we) begin

mem[addr] <= din;

end

end

assign dout = mem[addr];

endmodule

[Slice count is same]

[RTL shematic is same]

Single Port BRAM (50k bits LUT)

//Verilog code

```
`timescale 1ns / 1ps
```

```
module singlePORT_50kbits(
```

```
input clk, we, [10:0] addr, [31:0] din, output [31:0] dout
```

```
);
```

```
reg [31:0]mem [0:1562];
```

```
always @(posedge clk)
```

```
begin
```

```
if(we) begin
```

```
    mem[addr] <= din;
```

```
end
```

```
end
```

```
assign dout = mem[addr];
```

```
endmodule
```

1. Slice Logic

Site Type	Used	Fixed	Available	Util%
Slice LUTs*	937	0	53200	1.76
LUT as Logic	105	0	53200	0.20
LUT as Memory	832	0	17400	4.78
LUT as Distributed RAM	832	0		
LUT as Shift Register	0	0		
Slice Registers	0	0	106400	0.00
Register as Flip Flop	0	0	106400	0.00
Register as Latch	0	0	106400	0.00
F7 Muxes	384	0	26600	1.44
F8 Muxes	192	0	13300	1.44

RTL analysis schematic:

